

FOC Motor Controller — Rev C Action Summary

Consolidated design review, calculations, and pre-layout checklist

Target motor: ODrive M8325s 100KV | Gate driver: DRV8323S | MCU: STM32G431C8U6

0. How to use this sheet

Each section states the decision or finding, then lists action items with checkboxes. Work top-down: Section 1 (corrections to earlier advice) changes numbers used elsewhere, so read it first. Priority key: CRIT = blocks power-on; HIGH = blocks layout; MED = should be in Rev C; LOW = optional / next revision.

1. Corrections to Earlier Guidance (read first)

1.1 Charge pump current was understated

Earlier in this project I stated the DRV8323 charge pump supplies roughly 10-15 mA at a 50 V bus. That was wrong. The DRV8323 datasheet states the charge pump regulates to $V_{VM} + 11\text{ V}$ and supports an average output current of 25 mA. The lower figures (15 mA at $V_{VM} = 8\text{ V}$, 10 mA at 6 V) apply only at low bus voltages where the doubler is strained. At a 44-50 V bus, use $I_{VCP} = 25\text{ mA}$.

Consequence: the gate-charge ceiling roughly doubles versus what I previously told you. The IPT015N10N5 is less marginal than I implied, and 40 kHz becomes a realistic design point with the right MOSFET.

1.2 LMR38010FDDAR input rating

You confirmed from the datasheet that this part supports 80 V input. The earlier flag raised against it is withdrawn - no change needed for a 60 V bus. No action required.

1.3 Values still marked as estimates

- Q_g figures for ISC022N10NM6, ISC030N10NM6, ISC060N10NM6 and the BSC-series parts were scaled from the confirmed ISC027N10NM6 value (58 nC). They are estimates and must be read off each datasheet before ordering.
- M8325s phase resistance, phase inductance, and current rating are not published by ODrive. Every current/torque/power figure in this sheet is inverter-limited, not motor-verified.
- $K_t = 8.27 / KV = 0.0827\text{ Nm/A}$ is a first-order estimate from the KV rating, not a measured value.

2. Design Point and Derived Requirements

2.1 Motor and bus (published / derived)

Parameter	Value	Source
KV	100 RPM/V	ODrive published
Pole pairs	20	ODrive published
Diameter / weight	92 mm / 840 g	ODrive published
Kt (estimate)	0.0827 Nm/A	8.27 / KV - first order
Phase R / L	NOT PUBLISHED - measure	must characterise
Bus (12S LiPo)	42.0 V cutoff / 44.4 V nom / 50.4 V full	12 x 3.5/3.7/4.2 V
V_phase,rms max (SVPWM)	20.6 V at 50.4 V bus	$V_{bus} / (\sqrt{3} \times \sqrt{2})$

2.2 Switching frequency - driven by 20 pole pairs

$f_{elec} = (RPM / 60) \times 20$. At 1000 rpm = 333 Hz; 2000 rpm = 667 Hz; 3500 rpm = 1167 Hz. Using a rule of thumb of at least 20 PWM cycles per electrical cycle, full speed needs $f_{PWM} \geq 25$ kHz. Design target 40 kHz for headroom, ultrasonic operation, and lower current ripple.

2.3 Gate charge ceiling (corrected)

Sinusoidal 180 deg commutation, DRV8323 datasheet Eq. 10: $I_{VCP} > 3 \times Q_g \times f_{PWM}$

$Q_{g_max} = I_{VCP} / (3 \times f_{PWM})$, with $I_{VCP} = 25$ mA:

20 kHz -> 417 nC 30 kHz -> 278 nC 40 kHz -> 208 nC 45 kHz -> 185 nC

Design target with 2x margin at 40 kHz: $Q_g \leq 100$ nC

TI calls this a quick calculation, so do not design to the edge. A 200 nC part works at 40 kHz on typical silicon but sits at the limit on worst-case Q_g .

2.4 Current target and the loss inversion

To make full use of this motor, target ODrive S1/Pro class current: 40 A RMS continuous, 80 A peak. Loss per MOSFET at 40 A RMS, 40 kHz, 50 V bus ($R_{ds(on)}$ derated 1.5x for 125 C junction, ~50% duty, $I_{DRIVE} = 1$ A):

$R_{ds(on)} = 2.24$ mOhm -> $P_{cond} 2.69$ W + $P_{sw} 1.58$ W = 4.27 W $R_{ds(on)} = 2.70$ mOhm -> $P_{cond} 3.24$ W + $P_{sw} 1.36$ W = 4.60 W $R_{ds(on)} = 1.50$ mOhm -> $P_{cond} 1.80$ W + $P_{sw} 3.85$ W = 5.65 W (IPT015N10N5)

Note the inversion: the ultra-low- $R_{ds(on)}$ IPT015N10N5 has the WORST total loss at 40 kHz because its Q_{gd} is 2-3x higher. Switching loss dominates conduction loss at this current and frequency. Select on Figure of Merit ($R_{ds(on)} \times Q_g$), lower is better - not on $R_{ds(on)}$ alone.

Done	Action item	Priority	Where
[]	Fix f_{PWM} in firmware/CubeMX at the chosen value (40 kHz target) and record it on the schematic	HIGH	TIM1 / schematic note
[]	Measure M8325s phase resistance (4-wire, line-to-line / 2) and phase inductance (LCR)	HIGH	bench, pre-tuning
[]	Verify Kt empirically once the motor spins; do not rely on 8.27/KV for final torque claims	MED	bench
[]	Write the Q_g ceiling calculation as a text note next to the MOSFETs on the schematic	MED	MOSFETs sheet

3. MOSFET Selection

3.1 Required specification (derived in Section 2)

Parameter	Requirement	Rationale
V_DS	≥ 100 V	50.4 V bus x ~2 for switching transients
Q_g total @10V	≤ 100 nC target; 208 nC hard limit	Charge pump budget at 40 kHz
Q_gd (Miller)	as low as possible, ≤ 20 nC	Dominates switching loss
R_DS(on) @10V	2 - 3 mOhm	Conduction loss at 40 A RMS
I_D continuous	≥ 80 A (Tc)	Peak current headroom
Package	SuperSO8 5x6 or TOLL	Exposed pad for via-in-pad thermal path
V_GS(th)	≥ 2 V	dV/dt parasitic turn-on immunity

3.2 Candidates - Tier 1, Infineon OptiMOS 6 (best fit)

Part	V_DS	Rds(on)	Qg	I_D	Package	Availability
ISC022N10NM6	100V	2.24 mOhm max	~70 nC *	230 A	SuperSO8	Infineon / DK
ISC027N10NM6	100V	2.7 mOhm max	58 nC (conf.)	192 A	SuperSO8	Infineon / DK
ISC030N10NM6	100V	3.0 mOhm	~52 nC *	179 A (Tc)	PG-TDSON-8 FL	DigiKey, in stock
ISC060N10NM6	100V	6.0 mOhm max	~30 nC *	97 A	SuperSO8	DigiKey ~\$3.10

* estimated by scaling from the confirmed ISC027N10NM6 figure - verify from datasheet.

3.3 Candidates - Tier 2, OptiMOS 5 (confirmed DigiKey.ca stock)

Part	V_DS	Rds(on)	Qg	Package	Link
BSC040N10NS5	100V	4.0 mOhm	~60 nC *	PG-TDSON-8-7, 100 A (Tc)	DigiKey.ca
BSC027N10NS5	100V	2.7 mOhm	~90 nC *	SuperSO8	Infineon / DigiKey

3.4 Candidates - Tier 3, JLCPCB stock (all miss the Qg target)

Part	V_DS	Rds(on)	Qg	JLCPCB	Verdict
GT016N10TL	100V	1.3 mOhm	165 nC	C28405277	40 kHz marginal
NCEP023N10LL	100V	2.3 mOhm	240 nC @50V	C2686686	≤ 30 kHz only
NCEP020N10LL	100V	2.0 mOhm	252 nC	C2686687	≤ 30 kHz only
IPT015N10N5 (current)	100V	1.5 mOhm	169 typ / 211 max	-	40 kHz at the limit

The JLCPCB 100 V TOLL inventory is optimised for 200-360 A battery/BMS applications; none of it has the low-Qg profile this design wants.

3.5 Recommendation

- **Primary: ISC022N10NM6.** Best FOM in the list (2.24 mOhm x ~70 nC ~ 157 mOhm.nC, versus 253 for the IPT015N10N5). About 4.3 W per device at 40 A RMS / 40 kHz, with roughly 3x margin on the Qg ceiling.
- **Cost-optimised: ISC030N10NM6.** Confirmed DigiKey stock, very low Qg, slightly higher conduction loss and lower switching loss.
- **If restricted to JLCPCB:** keep the IPT015N10N5 and cap f_PWM at 30 kHz. It works, but you give up roughly the top 20% of the motor's speed range for clean current control.

Done	Action item	Priority	Where
[]	Read actual Qg and Qgd off the datasheet for the chosen part - do not use the * estimates	CRIT	datasheet
[]	Confirm the SuperSO8 / PG-TDSON-8-FL land pattern in KiCAD; FL (enlarged source) differs from standard TDSON-8	CRIT	footprint lib
[]	Update MOSFET symbol value, footprint, MPN and supplier fields in schematic + BOM	CRIT	MOSFETs sheet
[]	Re-run the Qg ceiling check for the final part at the final f_PWM	HIGH	calc
[]	Recompute gate resistor from the new part's Qgd and target rise time	HIGH	MOSFETs sheet
[]	Recompute snubber R/C from the new part's Coss curve at 50 V	HIGH	MOSFETs sheet

4. Power Stage Consequences of the 40 A Target

4.1 Shunt resistors are now wrong

At 40 A RMS the existing 5 mOhm 3 W shunts dissipate $I^2 \times R = 1600 \times 0.005 = 8 \text{ W}$ each - far over rating. Resize to 1 mOhm ($P = 1.6 \text{ W}$ at 40 A) and set the DRV8323 CSA gain to 20 V/V for a roughly +/-70 A range.

4.2 Thermal path is now mandatory, not optional

6 x ~4.3 W = ~26 W at 40 A continuous. This is not survivable on bare PCB copper. A chassis heatsink contact is required at this current level. Confirm the mechanical design provides a defined flat contact area on the board back beneath the MOSFET field, and specify the thermal interface material.

4.3 Realistic output

Torque at 40 A: $0.0827 \times 40 = 3.3 \text{ Nm}$. Torque at 80 A peak: $0.0827 \times 80 = 6.6 \text{ Nm}$. These follow from the estimated Kt and are inverter-limited, not motor-verified.

Done	Action item	Priority	Where
[]	Change shunt resistors 5 mOhm -> 1 mOhm; verify package power rating at 1.6 W with margin	CRIT	MOSFETs sheet
[]	Set DRV8323 CSA gain to 20 V/V in SPI init; verify ADC full-scale maps to the intended current range	CRIT	firmware + calc
[]	Recheck VDS_LVL overcurrent threshold against new Rds(on) and new trip current	HIGH	firmware
[]	Define chassis contact area and TIM in the mechanical drawing	HIGH	mechanical
[]	Re-verify phase solder pad and trace sizing per IPC-2221 at the new current	HIGH	layout
[]	Re-verify bulk capacitance ripple-current rating at the new phase current	HIGH	calc

5. Gate Driver - DRV8323S

5.1 Part number

The part number drawn as DRV8323SRTAR does not match a TI ordering code. For SPI control with no integrated buck in the 32-pin WQFN, the correct code is **DRV8323SRTVR** (RTV package). Update the symbol value, footprint and BOM.

5.2 Dual pin labels are correct as drawn

MODE_SDO / IDRIVE_SDI / VDS_SCLK / GAIN_SCS_N are TI's own dual-function notation (hardware-config name / SPI name). Your electrical connections (DRV_MISO -> MODE_SDO, DRV_MOSI -> IDRIVE_SDI, DRV_SCLK -> VDS_SCLK, DRV_CS -> GAIN_SCS_N) are correct for the S variant. No change needed to the labels themselves.

5.3 External component values against the datasheet

Pin / net	Datasheet requirement	Status
VM to VCP	1 uF, 25 V, X5R/X7R (charge pump storage)	verify
CPH to CPL	47 nF, X5R/X7R, VM-rated (flying cap)	raise voltage rating
DVDD to AGND	1 uF , X5R/X7R, 6.3 V	currently 100 nF - change
VM local bulk	>= 10 uF at the VM pin	currently 4.7 uF - change
VREF	0.1 uF to AGND	verify present
nFAULT, SDO	external pull-up (open-drain outputs)	verify both, and only these

Done	Action item	Priority	Where
[]	Change part number to DRV8323SRTVR; confirm RTV 32-pin WQFN footprint and exposed pad	CRIT	Motor Driver sheet
[]	Change DVDD cap 100 nF -> 1 uF X5R/X7R 6.3 V	CRIT	Motor Driver sheet
[]	Add >= 10 uF VM-rated ceramic directly at the DRV8323 VM pin (separate from stage bulk)	CRIT	Motor Driver sheet
[]	Raise CPH/CPL 47 nF cap voltage rating - TI rule is 2x margin over bus (>=120 V at 60 V bus)	CRIT	Motor Driver sheet
[]	Confirm 1 uF 25 V cap present between VM and VCP	HIGH	Motor Driver sheet
[]	Confirm pull-ups exist on nFAULT and SDO only - none on SCLK/MOSI/nSCS	HIGH	Motor Driver sheet
[]	Confirm no leftover hardware-config resistor dividers on IDRIVE/VDS/GAIN/MODE pins	HIGH	Motor Driver sheet
[]	Confirm 0.1 uF on VREF to AGND	HIGH	Motor Driver sheet

6. Microcontroller - STM32G431CBU6

6.1 Package check

Earlier review noted the symbol appearing as STM32G431CBT6 (LQFP-48) while the intended part is CBU6 (UFQFPN-48). These are different footprints. Confirm which one you are actually building and that symbol, footprint and BOM all agree.

6.2 Items to verify in CubeMX

- PA6 assigned to TIM1_BKIN (AF6), not SPI1_MISO (AF5). Break polarity must be **Active Low** - nFAULT is open-drain active-low. Wrong polarity means a fault asserts PWM on rather than off.
- ADC1 and ADC2 in Dual Regular Simultaneous mode, ADC1 master triggered from TIM1_TRGO2 at the center-aligned counter event, so the two phase currents are sampled at the same instant.
- ADC sampling window sits at least ~200 ns after the last switching edge.
- PB4 and PB6 carry a UCPD dead-battery pull-down (5.1 k) that is enabled by default. If these pins are used for SPI1_MISO / DRV_CS, add PWR->CR3 |= PWR_CR3_UCPD_DBDIS early in init.
- FDCAN1 on PA11/PA12 - no conflict because the native USB peripheral is unused (CH340X handles USB).

Done	Action item	Priority	Where
[]	Confirm MCU package: CBU6 (UFQFPN-48) vs CBT6 (LQFP-48) - symbol, footprint and BOM must agree	CRIT	MCU sheet
[]	Verify PA6 = TIM1_BKIN AF6 and Break polarity = Active Low	CRIT	CubeMX
[]	Verify nFAULT is physically routed to the TIM1_BKIN pin	CRIT	MCU sheet
[]	Configure ADC1/ADC2 dual simultaneous, trigger TIM1_TRGO2	CRIT	CubeMX
[]	Add PWR_CR3_UCPD_DBDIS write to early firmware init	HIGH	firmware
[]	Assign or No-Connect every unused MCU pin; re-run ERC after	HIGH	MCU sheet
[]	Set user labels on every net in CubeMX so generated HAL defines are readable	LOW	CubeMX

7. Remaining Board-Level Items

Done	Action item	Priority	Where
[]	Change battery-sense clamp from 1N4148 signal diode to MM3Z3V3T1G zener, cathode to sense node	HIGH	Battery sensing
[]	Change LED series resistors 100R -> 220R/330R (100R drives ~28 mA through 0603 LEDs)	HIGH	User LEDs
[]	Upgrade gate-drive parallel diodes 1N4148 -> BAT54S / PMEG2010 (1N4148 marginal at 1 A I_DRIVE)	HIGH	MOSFETs sheet
[]	Set snubber resistors to 1/4 W minimum; recompute $P = C \times V^2 \times f$ at final f_{PWM}	HIGH	MOSFETs sheet
[]	Recompute bulk capacitance with MLCC DC-bias derating at 50 V, not nominal capacitance	HIGH	MOSFETs sheet
[]	Add reverse-polarity protection at XT60 input (P-channel ideal diode, e.g. SI7157DP)	MED	Power Input
[]	Add ferrite bead between power connector and bulk capacitor bank	MED	Power Input
[]	Add common-mode choke on CAN_H/CAN_L at the connector end	MED	CAN FD sheet
[]	Add CAN node-ID selection hardware (DIP switch or 3x solder jumpers + pull-downs)	MED	Interconnects
[]	Confirm AUX 7-pin JST-GH pull-ups and decoupling present (10k on IO lines, 100 nF on 5V and 3V3)	MED	Interconnects

Done	Action item	Priority	Where
[]	Resolve every empty footprint library link before generating fab output	HIGH	all sheets
[]	Fill in title block fields (board name, project, revision, date, designer) on all sheets	LOW	all sheets
[]	Update the revision history sheet with the Rev B -> Rev C change log	LOW	Rev history

8. Calculations To Complete (formulas only - values are yours)

These use your final chosen MOSFET's real datasheet numbers. Record each result as a text note on the relevant schematic sheet - that is what turns component choices into a reviewable engineering design.

8.1 Charge pump / gate drive capacity

$$I_{VCP} > 3 \times Q_g \times f_{PWM} \text{ (sinusoidal, DRV8323 Eq. 10; } I_{VCP} = 25 \text{ mA)}$$

8.2 Gate resistor sizing

$$t_{rise} \sim Q_{gd} / I_{gate} \quad I_{gate} = V_{drive} / (R_{gate_ext} + R_{gate_internal})$$

8.3 Loss and thermal

$$P_{cond} = I_{rms}^2 \times R_{ds(on)_hot} \times duty \quad P_{sw} = 0.5 \times V_{bus} \times I_{pk} \times (t_r + t_f) \times f_{PWM} \quad P_{gate} = Q_g \times V_{gs} \times f_{PWM} \times 3 \quad T_j = T_{amb} + P_{dev} \times (R_{th_jc} + R_{th_iface} + R_{th_sink-amb}) \quad \text{target } T_j \leq 125 \text{ C}$$

8.4 Snubber

$$f_{ring} = 1 / (2\pi \sqrt{L_{loop} \times C_{oss}}) \quad R_{snub} = \sqrt{L_{loop} / C_{oss}} \quad C_{snub} = 1x..3x \ C_{oss} \quad P_{snub} = C_{snub} \times V_{bus}^2 \times f_{PWM} \text{ (resistor power rating)} \quad \text{Read } C_{oss} \text{ from the datasheet curve at 50 V, not the 25 V test figure.}$$

8.5 Miller / shoot-through margin

$$dV_{induced} = [C_{gd} / (C_{gd} + C_{gs})] \times (dV/dt) \times R_{gate_off} \text{ must stay well below } V_{GS(th)}$$

8.6 Bulk capacitance

$$C_{eff} = \sum (C_{nom_i} \times derating_i(V_{bias})) \quad \text{use each cap's DC-bias curve} \quad C_{min} = I_{ripple_pk} / (f_{PWM} \times dV_{allowed}) \quad \text{Also check the bank's ripple-current (A RMS) rating at worst-case modulation index.}$$

8.7 Motor characterisation (bench)

Phase R: 4-wire line-to-line measurement / 2 for wye per-phase. Phase L: LCR meter at current-loop bandwidth, or locked-rotor voltage step. $f_{elec} = (RPM / 60) \times 2\pi \rightarrow$ cross-check against chosen f_{PWM} and ADC rate.

Done	Action item	Priority	Where
[]	Complete 8.1 - 8.6 with final part datasheet values and annotate results on the schematic	HIGH	all
[]	Build LTspice half-bridge model with the manufacturer SPICE model; validate 8.2, 8.4, 8.5	HIGH	pre-layout
[]	Complete 8.7 on the bench before tuning current-loop gains	HIGH	bench

9. Gate Checklist Before Moving to Layout

Done	Action item	Priority	Where
[]	All CRIT items in Sections 3-6 closed	GATE	-
[]	ERC clean, with only deliberately-suppressed benign warnings remaining	GATE	-
[]	Every component has a resolved footprint and a real orderable MPN	GATE	-
[]	Final MOSFET chosen, ordered or stock-confirmed, footprint verified against datasheet drawing	GATE	-
[]	f_{PWM} decided and consistent across firmware, Q_g check, snubber and filter calculations	GATE	-

Done	Action item	Priority	Where
[]	Shunt value, CSA gain and OCP threshold all consistent with the final current target	GATE	-
[]	Thermal solution defined (chassis contact area, TIM) and reflected in the mechanical drawing	GATE	-
[]	LTspice half-bridge simulation run and results recorded	GATE	-

End of summary. Figures marked as estimates must be replaced with datasheet or measured values before fabrication.